

Presence in VR: Developing measure and stimulus

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ABSTRACT

This is a research project that involves development and further validation of the VR application that can be used by researchers as a behavioral measure of the sense of presence in VR, more specifically, of the factors unrelated to the virtual content. Another side of the project relates to one of the possible factors, which is priming via preamble, or a virtual replica of the immediate physical environment, used to stimulate presence. It is planned to manipulate the realisticness of the preamble, where less realistic preamble can lower expectations towards the next virtual environment increasing chances for the sense of presence to appear. The project is on the 'pre-research stage' due to the difficulties that the pandemic has brought in many aspects of our life. Therefore, one of the questions that is necessary to discuss is how to conduct a subject-based research at the times of 'social' distancing.

Index Terms: Presence—Virtual Reality—Behavioral measure—Priming

1 INTRODUCTION

VR might be a powerful tool for simulating experiences that for some reason we cannot obtain in real life. But in order to simulate those experiences we need achieve or feel presence, which is a perceptual illusion of non-mediation [4] or non-simulation, i.e. a psychological state due to which the role of technology is left unnoticed in our experiences. The effects of presence are especially explicit within such fields as VR therapy [6], entertainment [8], online education [5], and so on. Therefore, to study those effects or to study how immersive VR can be, it is necessary to consider and study presence. Even though presence is related to other media technologies, VR is the field from which the concept of presence (along with immersion) emerged. As a perceptual illusion of non-mediation/simulation, presence depends on many factors where the content of the VEs (virtual environments) and technology are among many things that can shape cognitive processes and therefore promote presence. VEs are also affordable for measuring presence behaviorally, which I prefer to self-reports that are frequently applied in the scholarship.

2 PRESENCE SCHOLARSHIP

This project focuses on spatial or, rather, physical presence that is often associated with VR. Despite the fact that presence is a psychological/cognitive state, many researchers concentrate on the technological factors of presence, which are still important but they do not constitute the whole variety of ways by which this perceptual illusion can formulate. And even though presence scholarship includes operationalizable definitions of presence [4], there is still a conceptual ambiguity [3] in the field. Along with others issues, it also hinders valid measurement of presence. While there are many ways to measure presence, self-reports are still a dominant way to do so. These is a broad rationale behind my project.

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2.1 Behavioral measure of presence: VR app

Depending on the VE design, presence can be measured via behavioral observations of users', for instance, bending, head moving, swaying, ducking or via other postural change [3, 10]. It is possible to measure presence this way because of the principle of 'behavioral realism' [3]. In other words, when we feel that a virtual environment (or some part of it) is non-mediated or non-simulated we can act accordingly. As a result, currently, I am developing VEs that afford behavioral response as a 'realistic' reaction which indicates the state of presence.

Quite often, VR researchers and practitioners use threatening environments to stimulate behavioral response. In fact, researchers indicated that anxiety functions as a moderator [2], i.e. enhancing the state of presence for the VR users. This will be a main motivator in my measure as well.

Based on my previous pilot-study with a moving ceiling, I am developing VE(s) via Unity3D to stimulate and assess behavioral response via several but simple reactions or body movements within certain axes (i.e. X, Y, and/or Z). The stimuli will include moving/approaching elements of the room, as in wall and ceiling parts. In addition, I can replicate the situation of the height experiments with a pit situation to measure the rate of the users' reaction. I am planning to test several environments. One of the independent variables will be the level of social realism, or correspondence with the social reality. In other words, one setting will be similar to the real life circumstances, while another will represent some fictional setting, such as a sci-fi space location.

At the same time, I am considering whether it is useful or not to include some task in the VEs because it might help users to concentrate on something and stay focused. But, it also can be too disturbing, which can possibly relocate their attention from the behavioral stimulus. In addition, noticeability of the stimuli is a also a matter of concern.

In the end, I hope to create accessible and reliable measure, specially a VR app, that can be used by VR researchers and practitioners to test factors of presence in order to make their VEs more immersive.

2.2 Realism of the preamble as a factor of presence

I am also planning to develop VEs which I will later use to stimulate presence. Overall, I am interested in various non-content (unrelated to VEs) factors that can constitute presence and/or shape perceptual process. It may be different version of HMDs, previous experience with technology [3], individuals' mood, the level of surrounding noise, willingness to experiences presence [7], strong distracting smell [1] and so on. There are multiple ways to make VR a more immersive experience for its users.

One of the contextual ways to stimulate presence is a preamble, i.e. "a virtual replica of the immediate physical environment, having the sole purpose of giving a user a more natural transition into VR, rather than immediately throwing them into the infinite possibilities of a VR experience" [11, p.154]. Preamble represents a sort of priming that can be used to contextually enhance immersive experiences in VR despite the content of the VE or a technology used to access it.

In order to test the preamble in a more advanced manner, I am planning to manipulate the level of its realistic-ness or detail-ness (as in basic 3D model vs highly realistic replica). The hypothesis is

that when a user transitions to VR, lower quality of the simulated preamble can lower expectations towards the next virtual simulation and its circumstances. Therefore, lower expectations can shift cognitive processing of reality. Expectations are critical, they should be met to gain presence [9]. Otherwise, it can break presence.

As a result, it can be true that simple reproduction of the external environment (made instantly via photogrammetry, for instance) can be a more effective preamble to immerse into VEs than a detailed virtual 3D-modelled one. Although I still need to validate the VR app as behavioral presence measure to actually use it to test presence factors, the project with a preamble in itself can be significant for the field.

3 DISCUSSION: AREAS FOR IMPROVEMENT

There are several issues I wanted to discuss. First one is related to the social research per se and can be relevant to many of us. Other questions are more specifically related to the project of mine.

3.1 Social research during pandemic

One of the key questions that I would like to discuss is relevant for a lot of researchers in social sciences. As I already noted, it seems like it will be quite difficult, at least in the nearest future, to conduct subject-based research. Therefore, my question will be – how can (especially young) scholars overcome those obstacles? It is true that creating an application that can be used by the participants independently [12] is potentially a viable way to conduct studies, even without incorporating questionnaires into it, but, this method still lacks control. As well as, there are issues with accessibility of the technology, platform-dependence and so on.

3.2 Factors of presence/immersion in(to) VR

Considering that VR practitioners within the industry are dominantly interested in technological factors of presence/immersion, how is it possible to promote all the other ways presence can appear from or be stimulated with? Even though presence is a result of interaction with media technology, it is still a psychological/cognitive state.

3.3 Measuring presence behaviorally

Finally, I am curious if someone is also thinking about developing VEs to measure presence. Specifically, how would anyone stimulate ‘natural’ behavioral reaction to the virtual circumstances, objects, or other features? Even though fear/anxiety is considered to be an appropriate stimulator/motivator, is there a fundamentally different way to do it? Also, as results of my pilot study shows, some users responded to the stimulus because they thought it was a game. How to avoid resemblance of animated VEs to video games, which motivates users to participate but hinders ‘natural’ reaction?

4 CONCLUSION

Overall, this project focuses on conceptual and methodological advancement of presence studies, and specifically of physical presence in VR. It is crucial to have clear conceptualization of the phenomena and valid measures to move forward as a research field. At the same time, even though presence depends on technology, and is a result of interaction with media/simulation technology, it is still a psychological state that innately depends on perceptual process, including what and how is something perceived. And finally, I hope that social sciences dealing with subjects will adapt to the new circumstances we are all facing these days.

REFERENCES

- [1] Baus, O., Bouchard, S. Exposure to an unpleasant odour increases the sense of presence in virtual reality. *Virtual Reality*, 21(2):59-74, 2017. doi:10.1007/s10055-016-0299-3
- [2] Bouchard, S., St-Jacques, J., Robillard, G., Renaud, P. Anxiety increases the feeling of presence in virtual reality. *Presence: Teleoperators and Virtual Environments*, 17(4):376-391, 2008. doi:10.1162/pres.17.4.376
- [3] Freeman, J., Avons, S. E., Meddis, R., Pearson, D. E., IJsselstein, W. Using behavioral realism to estimate presence: A study of the utility of postural responses to motion stimuli. *Presence: Teleoperators Virtual Environments*, 9(2):149-164, 2000. doi:10.1162/105474600566691
- [4] Lombard, M., Ditton, T. At the heart of it all: The concept of presence. *Journal of computer-mediated communication*, 3(2), JCMC321, 1997. doi:10.1111/j.1083-6101.1997.tb00072.x
- [5] Mikropoulos, T. A., Strouboulis, V. Factors that influence presence in educational virtual environments. *CyberPsychology Behavior*, 7(5):582-591, 2004. doi:10.1089/cpb.2004.7.582
- [6] Parsons, T. D., Rizzo, A. A. Affective outcomes of virtual reality exposure therapy for anxiety and specific phobias: A meta-analysis. *Journal of behavior therapy and experimental psychiatry*, 39(3):250-261, 2008. doi:10.1016/j.jbtep.2007.07.007
- [7] Sas, C., O'Hare, G. M. Presence equation: An investigation into cognitive factors underlying presence. *Presence: Teleoperators Virtual Environments*, 12(5):523-537, 2003. doi:10.1162/105474603322761315
- [8] Skalski, P., Denny, J., Shelton, A. Telepresence and media effects research. In: Bracken, CC. Skalski, P. *Immersed in media: Telepresence in everyday life*, Routledge, , 2010, 158-180.
- [9] Slater, M., Lotto, B., Arnold, M. M., Sánchez-Vives, M. V. How we experience immersive virtual environments: the concept of presence and its measurement. *Anuario de Psicología*, 40:193-210, 2009.
- [10] Slater, M., McCarthy, J., Maringelli, F. The influence of body movement on subjective presence in virtual environments. *Human factors*, 40(3):469-477, 1998. doi:10.1518/001872098779591368
- [11] Smolentsev, A., Cornick, J. E., Blascovich, J. Using a preamble to increase presence in digital virtual environments. *Virtual Reality*, 21(3):153-164, 2017. doi:10.1007/s10055-017-0305-4
- [12] Steed, A., Frlston, S., Lopez, M. M., Drummond, J., Pan, Y., Swapp, D. An ‘in the wild’ experiment on presence and embodiment using consumer virtual reality equipment. *IEEE transactions on visualization and computer graphics*, 22(4):1406-1414, 2016. doi:10.1109/TVCG.2016.2518135